

Use SCP to securely transfer files between two Unix computers

In Unix (agat), you can use SCP (the `scp` command) to securely copy files and directories between remote hosts without starting an FTP (aerg) session or logging into the remote systems explicitly. The `scp` command uses SSH (aelc) to transfer data, so it requires a password or passphrase for authentication (alqk). Unlike `rcp` or FTP, `scp` encrypts both the file and any passwords exchanged so that anyone snooping on the network cannot view them.

Note:

Two-factor authentication using Two-Step Login (Duo) (beum) is required for access to the login nodes on IU research supercomputers (alde), and for SCP and SFTP file transfers to those systems. SSH public key authentication remains an option for researchers who submit the "SSH public key authentication to HPS systems" agreement  (<https://hpceverywhere.iu.edu/agree#/agree>) (log into HPC everywhere (aquv) using your IU username and passphrase (beml)), in which you agree to set a passphrase on your private key when you generate your key pair. If you have questions about how two-factor authentication may impact your workflows, contact the UITS Research Applications and Deep Learning team  (<https://mailform.kb.iu.edu/email.php?cid=1449>). For help, see Get started with Two-Step Login (Duo) at IU (bfgm) and Help for Two-Step Login (Duo) (aluu).

Syntax

The syntax for the `scp` command is:

```
scp [options] username1@source_host:directory1/filename1 username2@destination_host:directory2/filename2
```

The location of the source file is specified by `username1@source_host:directory1/filename1`, which includes the:

- Name of the account on the host computer (`username1`)
- Hostname of the computer on which the source file resides (`source_host`)
- Name of the directory containing the source file (`directory1`)
- Filename of the source file (`filename1`)

The location to which the source file will be copied is specified by `username2@destination_host:directory2/filename2`, which includes the:

- Name of the account on the destination computer (`username2`)
- Hostname of the computer to which the source file will be copied (`destination_host`)
- Name of the directory to which the source file will be copied (`directory2`)
- Filename of the copy (`filename2`)

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Make sure to include a space between the source and destination paths. Also, be careful when copying files that share the same name on both hosts; you may accidentally overwrite data you intended to keep.

For more about `scp`, consult its [manual page \(afjm\)](#). At the Unix prompt, enter:

```
man scp
```

At Indiana University, for personal or departmental Linux or Unix systems support, see [Get help for Linux or Unix at IU \(beqc\)](#).

Examples

For the following examples, assume your username is `dvader`, and you are logged into your account on the computer `empire.gov`:

- To copy a file called `rebels.txt` from your home directory on `empire.gov` to a directory called `revenge` in your account on the computer `deathstar.com`, enter:

```
scp ~/rebels.txt dvader@deathstar.com:~/revenge
```

You'll be prompted for your password on the destination system (`deathstar.com`). The command won't work unless you enter the correct password.

- To copy a directory (and all the files it contains), use `scp` with the `-r` option. This tells `scp` to recursively copy the source directory and its contents.

To copy the entire `revenge` directory from your `deathstar.com` account to your `empire.gov` account, enter:

```
scp -r dvader@deathstar.com:~/revenge ~/revenge
```

You'll be prompted for your password on the source system (`deathstar.com`). The command won't work unless you enter the correct password.

- To copy multiple files within a directory, you can use wildcards (for example, `*` or `?`). However, to use wildcards for copying multiple source files from a remote system, you need to place quotes (`" "`) around the path to the source files. This is necessary because the Unix [shell \(agvf\)](#), not the `scp` command, expands unquoted wildcards.

Therefore, to copy all the `.txt` files from the `revenge` directory on your `deathstar.com` account to your `revenge` directory on `empire.gov`, enter:

```
scp dvader@deathstar.com:"revenge/*.txt" ~/revenge/
```

You'll be prompted for your password on the source system (`deathstar.com`). The command won't work unless you enter the correct password.

For the following example, assume you (`dvader`) are logged into another computer (that is, some other computer that's not `empire.gov` or `deathstar.com`). To copy `luke.txt` from your home directory on `empire.gov` to your `revenge` directory on `deathstar.com`, enter:

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- `scp dvader@empire.gov:~/luke.txt dvader@deathstar.com:~/revenge`
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You'll be prompted to enter two passwords: one for the source system (`empire.gov`) and one for the destination system (`deathstar.com`). The command won't work unless you correctly enter both passwords.

Related documents

[Set up SSH public key authentication to connect to a remote system \(aews\).](#)

[About ssh-agent and ssh-add in Unix \(aeww\).](#)

This is document agye in the Knowledge Base.

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